BRYCE JOSEPH

(650) 452-0002 • brycejo@seas.upenn.edu
linkedin.com/in/brycejoseph • github.com/brycej217

EDUCATION

University of Pennsylvania, Philadelphia, PA — Aug 2025 - Dec 2026
Master of Science in Engineering, Computer Graphics and Game Technology | GPA: 3.7
Relevant Coursework: GPU Programming & Architecture, Interactive Computer Graphics, Computer Animation

New York University, New York, NY — Jan 2023 - May 2025
Bachelor of Arts, Computer Science and Mathematics | GPA: 3.5
Relevant Coursework: Algorithms, Differential Geometry, Topology, Object-Oriented Programming

TECHNICAL SKILLS

Programming Languages:	C++, C, Python, C#, GLSL/HLSL, TypeScript
GPU Programming:	CUDA, OpenCL, Nsight Systems/Compute
Graphics & Tools:	DX12, Vulkan, WebGPU, OpenGL, RenderDoc, Unity, Git

EXPERIENCE

Research Assistant: ***Occidental College,*** Los Angeles, CA — Sep 2022 - Dec 2022
- Contributed to research on emotion-driven facial animation system in Unity, proposing a shift from an RNN to reinforcement learning approach to improve generalization and responsiveness in unseen dialogue situations.

ETL Intern: ***Orbitax,*** San Francisco, CA — Jun 2022 - Aug 2022
- Built a production ETL pipeline between EY and Orbitax using Microsoft Azure Data Factory, Python, and custom APIs, automating ingestion and transformation of JSON data to enable seamless downstream analytics.

PROJECTS

glRemix (DX12, C++) — Nov 2025 - Dec 2025
DX12-based remastering platform that intercepts fixed-function OpenGL applications and renders them with real-time path tracing and asset replacement, without modding or source access.
- Developed the **OpenGL-to-DX12 driver**, translating OpenGL command streams into DXR path-tracing resources and operations, enabling titles like *Half-Life* and *Unreal* to run fully path traced at **up to 72 FPS**.
- Reimplemented **fixed-function OpenGL behavior** (geometry submission, state tracking, matrix stacks, materials, lighting, textures), converting an immediate-mode API into a modern DX12 resource and command model.

CUDA Path Tracer (CUDA, C++) — Sep 2025
GPU-accelerated Monte Carlo path tracer implemented entirely in CUDA.
- Optimized performance with **BVH traversal**, path termination, and memory coalescing, achieving **1300% speedup** on complex scenes.
- Added **physically-based rendering** features such as diffuse and specular BRDF sampling, normal and roughness mapping, depth of field, and environment mapping to achieve photorealistic global illumination.

CUDA Boids Flocking Simulation (CUDA, C++) — Sep 2025
Parallel flocking simulation using CUDA with spatial partitioning.
- Simulated **100,000+** boids with uniform and **coherent grid spatial partitioning** to reduce neighbor searches from $O(n^2)$ towards **$O(n)$**.
- Optimized **kernel occupancy** and memory locality with **Nsight** and increased throughput via coherent memory layouts and restricted neighbor-search regions.